中国科学技术大学

学术学位研究生学位论文开题报告

论文题目: Search for type III seesaw heavy leptons in pp collisions at $\sqrt{s} = 13.6$ TeV with the ATLAS detector

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所在院系: 近代物理系

学科专业: 粒子物理与原子核物理

研究方向: 实验粒子物理

填表日期: 2025 年 12 月 21 号

中国科学技术大学研究生院培养处

二零二五年三月　制表

说　明

1. 学位论文的开题报告是保证论文质量的一个重要环节，为了加强对研究生培养的过程管理，规范其学位论文的开题报告，特制此表。

2. 学位论文开题报告，应该在学位授予点或培养单位组织的学术报告会上报告，听取意见，论证后再填写此表。

3. 此表经导师签字后，交培养单位研究生教学管理办公室存档。

4. 研究生在提交学位申请时，必须提交该学位论文开题报告。

一、简况

<table>
<tr><td rowspan="3">研究生简况</td><td>姓　名</td><td>AHMED MEDHAT MOSTAFA ELSAYED</td><td>学　号</td><td>SL23004001</td><td>性　别</td><td>男</td></tr>
<tr><td>学生类别</td><td colspan="5">☑ 学术学位硕士　　☐ 学术学位博士</td></tr>
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<tr><td>英 文</td><td colspan="5">Search for type III seesaw heavy leptons in pp collisions at $\sqrt{s} = 13.6$ TeV with the ATLAS detector</td></tr>
<tr><td rowspan="5">论文摘要和关键词</td><td>摘要</td><td colspan="5">This study investigates the phenomenology of TeV-scale Type III Seesaw heavy leptons at the ATLAS detector, focusing on a distinctive signature where the daughter electroweak and Higgs bosons decay hadronically and are reconstructed as single large-R jets, rather than resolved jets. Dedicated analysis channels leveraging advanced jet substructure observables and multivariate analysis techniques are developed to maximize sensitivity to the SU(2) leptonic triplets in the TeV mass range. We project mass exclusion limits at 1.42 TeV and 1.30 TeV, at integrated luminosities of 400 fb^{-1} and 140 fb^{-1} respectively, with 95% confidence level.</td></tr>
<tr><td rowspan="3">关键词</td><td colspan="5">关键词数量三至五个，主题词之间空一格（英文用“ / ”分隔）</td></tr>
<tr><td>中文</td><td colspan="4">III 型跷跷板机制，大半径喷注，多变量分析技，ATLAS 探测器</td></tr>
<tr><td>英文</td><td colspan="4">Type III seesaw / large-R jets / multivariate analysis techniques/ ATLAS detector</td></tr>
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二、选题依据

The Standard Model (SM) of particle physics, despite its remarkable success in describing fundamental particles and their interactions, fails to explain the observed non-zero neutrino masses. Neutrino oscillation experiments have conclusively demonstrated that neutrinos have mass, yet the SM predicts them to be massless [1-3]. This discrepancy represents one of the most compelling pieces of evidence for physics beyond the Standard Model. Several theoretical frameworks have been proposed to explain the smallness of neutrino masses, with the seesaw mechanism being among the most elegant and well-motivated [4]. The type III seesaw (T3S) model extends the SM by an $SU(2)_L$ triplet of fermions that transform as $(1, \mathbf{3}, 0)$ under the SM guage group $SU(3)_c \times SU(2)_L \times U(1)_Y$, commonly refereed to as heavy lepton triplets. These triplets couple to the SM leptons and the Higgs field, generating neutrino mass via the seesaw mechanism after the electroweak symmetry breaking (EWSB).

At the LHC, the type III seesaw leptonic $SU(2)_L$ triplets are pair produced through electroweak (EW) interactions in both charged-charged (N^+N^-) and charged-neutral ($N^\pm N^0$) modes [4]. The heavy leptons are assumed to mix with SM leptons and decay to a $W^\pm$, Z or Higgs boson (H) and a SM lepton (ν or $\ell^\pm = e^\pm, \mu^\pm, \tau^\pm$). The branching ratio ($B_\ell$) for a heavy lepton decay into a light lepton of flavour ℓ is proportional to $|V_\ell|^2 / \sum_\alpha |V_\alpha|^2$, where V_ℓ is the corresponding mixing matrix element. Electroweak precision measurements enforce an upper limit on the mixing angles of 10^{-4} across all lepton flavours [5-6]. This bound is sufficiently large to ensure prompt decays of heavy leptons in the mass ranges accessible to current collider experiments. It also renders negligible effects of the radiative mass splitting between the neutral and charged heavy fermions [7-10]. In this study, the leptonic triplets are assumed to be degenerate in mass and their decays are assumed to be prompt. Their decay branching fractions to different SM bosons are determined solely by their masses. The free parameters are heavy leptons mass (m_N) and the decay branching fractions to the SM lepton flavours B_e, B_μ and B_τ, where we assume $B_e + B_\mu + B_\tau = 1$.

Leptonic decays of the daughter SM bosons result in a set of rich and unique multileptonic signatures that have low SM backgrounds. Consequently, previous searches by the LHC experiments have predominantly focused on the multileptonic final states, and the results are indeed encouraging [11-12]. The most stringent limits on the type III seesaw model come from a CMS search conducted in 2022 using the combined LHC dataset from 2016-2018 at $\sqrt{s} = 13$ TeV in multileptonic final states with up to four electrons and muons [12]. The search excluded at 95% confidence level (CL) type III seesaw fermions with masses below 980 GeV in the $B_e = B_\mu = B_\tau = 1/3$ mixing scenario and 990/1065/890 GeV in the exclusive electron/muon/tau coupling scenario.

For heavier triplet fermions, not only the extremely small pair production cross section of the triplet pairs further suppresses the sensitivity of the multileptonic signals but also the small branching fractions of the leptonically decaying SM bosons and the relatively larger SM backgrounds. Hence, to improve the sensitivity one has to consider the hadronic final states produced by hadronically decaying SM bosons, which are known to be embedded with large SM background contributions. The characteristic

signature of TeV scale triplets at the LHC would be vastly boosted final state lepton and SM boson that are likely to be largely separated in the azimuth plane. For the daughter SM bosons boosted to large p_Ts, their hadronic decay and fragmentation yield a collimated spray of hadrons which a standard jet algorithm would reconstruct as a single jet and thus are more likely to manifest as a **single large-R jet** rather than two resolved jets. Such phenomenology can only happen because of the large heavy leptons mass, that pushed the SM boson and lepton apart, and boosts the SM enough to reduce the ΔR between its daughter particles given their relatively smaller masses.

Previous phenomenological work have considered such final states [7], specifically in final states with 1J1L, 2J1L, 1J-2L-OS, 1J-2L-SS, 1J3L, 3L, and 4L. It is a cut-based analysis that use tagging techniques like N-subjetness for selecting the bosonic large-R jets. They report their method can probe triplet fermions with mass as large as 1265 (1380) and 1480 (1600) GeV with 5σ (3σ) significance at 500/fb and 3000\fb luminosities, respectively.

In this analysis, we consider final states that include multiple large-R jets and charged leptons. We use novel techniques for tagging the signal bosonic large-R jets and multivariate techniques to classify our signal events from SM background events. Results from every channel are then combined together and used to set expected exclusion limits on the triplet lepton masses using 140-1000/fb of pp collision luminosity.

References:

[1] Phys. Rev. Lett. 89 (1 June 2002), p. 011301.

[2] Phys. Rev. Lett. 87 (7 July 2001), p. 071301.

[3] Phys. Rev. Lett. 81 (8 Aug. 1998), pp. 1562–1567.

[4] Eur. Phys. J. C 72, 1899 (2012)

[5] J. High Energy Phys. 05 (2020) 022.

[6] Nucl. Phys. B966, 115374 (2021).

[7] J. High Energy Phys. 12 (2007) 061.

[8] Front. Phys. 6, 40 (2018).

[9] Phys. Lett. B 819, 136403 (2021).

[10] Phys. Lett. B 825, 136889 (2022).

[11] **CMS-2012** (10.1016/j.physletb.2012.10.070), **ATLAS-2015/8** (10.1103/PhysRevD.92.032001), **ATLAS-2015/9** (10.1007/JHEP09(2015)108), **CMS-2017** (10.1103/PhysRevLett.119.221802), **CMS-2020** (10.1007/JHEP03(2020)051), **ATLAS-2021** (10.1140/epjc/s10052-021-08929-9), **ATLAS-2022** (10.1140/epjc/s10052-022-10785-0)

[12] **CMS-2022** (10.1103/PhysRevD.105.112007)

三、研究的创新之处

a) **Targets ATLAS Run-3 data ($\sqrt{s} = 13.6$ TeV).** Extends previous searches performed at $\sqrt{s} = 13$ TeV and below by targeting the higher center-of-mass energy and larger Run-3 dataset, thereby increasing potential signal yield and discovery reach.

b) **Comprehensive multivariate analysis pipeline.** Rather than a single classifier, the analysis integrates multiple MVAs: constituent-level ML taggers for boosted boson identification and boosted decision trees for event-level signal discrimination. This layered MVA design lets each stage learn complementary features and reduces model reliance on any single observable.

c) **New discriminative variables for Type-III signatures.** We propose and test novel combinations of jet substructure observables (including N-subjettiness ratios and energy-correlation function derivatives) specifically optimized to separate the expected decay topologies of Type-III fermion triplets from dominant backgrounds.

d) **Constituent-level ML for fat-jet tagging.** Strong classifiers trained on particle-flow (constituent) inputs improve tagging performance for W/Z/H fat jets relative to taggers based solely on high-level variables. Their outputs are fed into the event-level BDTs to amplify discriminating power.

e) **Synergy of substructure observables and ML outputs.** By incorporating both engineered substructure features and ML-derived tagger scores into the final BDT, the analysis exploits orthogonal information channels (physics-motivated observables + data-driven classifiers) to maximize sensitivity

四、研究方法与技术路线

I. Event Generation and Object Reconstruction

For the purpose of this study, we consider all 27 distinct leading order (LO) production and decay processes resulting from the decays of each of N^+N^0 , N^+N^- and N^-N^0 pairs into nine different pairs of $W^\pm$, Z and H bosons. Since the neutral component N^0 is a Majorana neutrino, both the $N^0 \to l^+W^-$ and $N^0 \to l^-W^+$ channels are included with equal rates. Parton level events are generated with the **MadGraph5_aMC@NLO v3.5.5** framework using the **five-flavor scheme (5FS)** and the **NNPDF4.0_LO** ($lhaid = 331900$) parton distribution function (PDF) sets. The implemented model used for this analysis is given in [4]. The matrix element (ME) level events are interfaced to **PYTHIA 8.313** for bosonic decays, parton showering, fragmentation and hadronization, tuned with the **A14** parameter set and the **NNPDF2.3LO** PDF set. The ISR, FSR and MPI effects were also included. Different signal samples are generated for different triplet fermion mass hypothesizes in the range **1.0-2.0 TeV with a mass step of 50 GeV.**

We simulate and include all SM background processes giving rise to one or more prompt leptons and fatjets, such $t\bar{t}, t\bar{t} + V$ (assuming $m_t = 172.5$ GeV), **diboson ($VV, V = W^\pm, Z$), triboson (VVV), $V + nj$ and $t\bar{t} + nj$, with $n \leq 3$ (followed by MLM matching).** All the background events are generated by using the **sm-full** model of MadGraph5_aMC@NLO v3.5.5at LO using the 5FS followed

parton showering in PYTHIA 8.313. t-channels are set open for background process that require them. The background samples are normalized at least to the NLO prediction.

The signal and background MC events samples are passed through a fast simulation of the **ATLAS detector** response using the **Delphes** modular framework to reconstruct different objects, namely isolated electrons, muons, jets, photons and missing energy. **The soft-drop algorithm** *is applied with parameters* $\beta = 0$ *and symmetry cut* $z_{cut} = 0.1$ *, which systematically removes soft wide-angle radiation, preserving the hard core of the jet while improving the discrimination of boosted objects.*

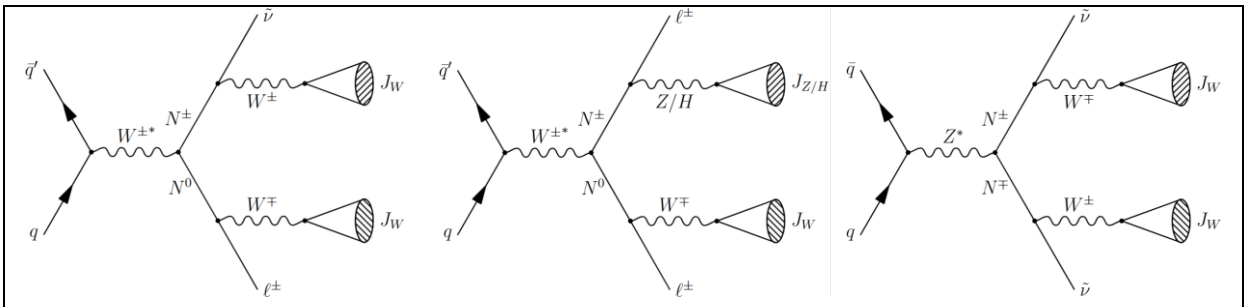
II. Event Selection

We split the signal final states into **seven** distinct and well-defined categories based primarily on the number of the light leptons. These eight channels are orthogonal, and are defined in the descending order of branching ratios as follows:

1) exactly **0** light leptons and **at least two** fatjets (2J0L),
2) *exactly **1** light lepton and **at least one** fatjet (1J1L),*
3) exactly **2 OS** light lepton and **at least one** fatjet (1J2L-0S),
4) exactly **2 SS** light lepton and **at least one** fatjet (1J2L-SS),
5) exactly **3** light lepton and **at least one** fatjet (1J3L),
6) exactly **3** light lepton and exactly **0** fatjets (0J3L),
7) at least **4** light leptons with **any number** of fatjets (4L)

At parton-level, final states including 2 light leptons mostly include two fatjets. Unfortunately, the jet reconstruction is not very efficient as the muon or electron reconstruction. Thus, the requirements of exactly two fatjets might increase the purity of the final state but at the same time rejects a significant number of signal events making the analysis difficult due to low statistics. For this reason, our selections include at least one fatjets and not precisely two. This allows the possibility for losing one fatjet during the reconstruction process. **We begin preforming the analysis with the 1J1L channel,** since it is expected to have the highest event yield. Below is a summary table describing our event selection:

Object	Requirement	Criteria
Lepton	Selection Relative Isolation	$p_T > 10$ GeV, $\lvert\eta\rvert \leq 2.5$ 12% ($e^{\pm}$), 15% ($\mu^{\pm}$) for $\Delta R \leq 0.5$.
Jet	Selection Clustering	$p_T > 10$ GeV, $\lvert\eta\rvert \leq 2.5$ Anti-k_T with $R_0 = 0.4$
Large-R jet	Selection Clustering Soft-Dropping	$p_T > 200$ GeV, $\lvert\eta\rvert \leq 2.5$ Anti-k_T with $R_0 = 1.0$ $\beta = 0,\ z_{cut} = 0.1$,
1J1L Event (Basic)	Number of Objects Requirements	$n(\ell^{\pm}) = 1,\ n(J) \geq 1$, $p_T(\ell^{\pm}) > 30$ GeV, $p_T(J_{SD}) > 200$ GeV $\min \Delta R(\ell^{\pm}, J_{SD}) \geq 0.6$,



Example diagrams for 1J2L, 2L2J, and 012J processes, in descending order of total cross section.

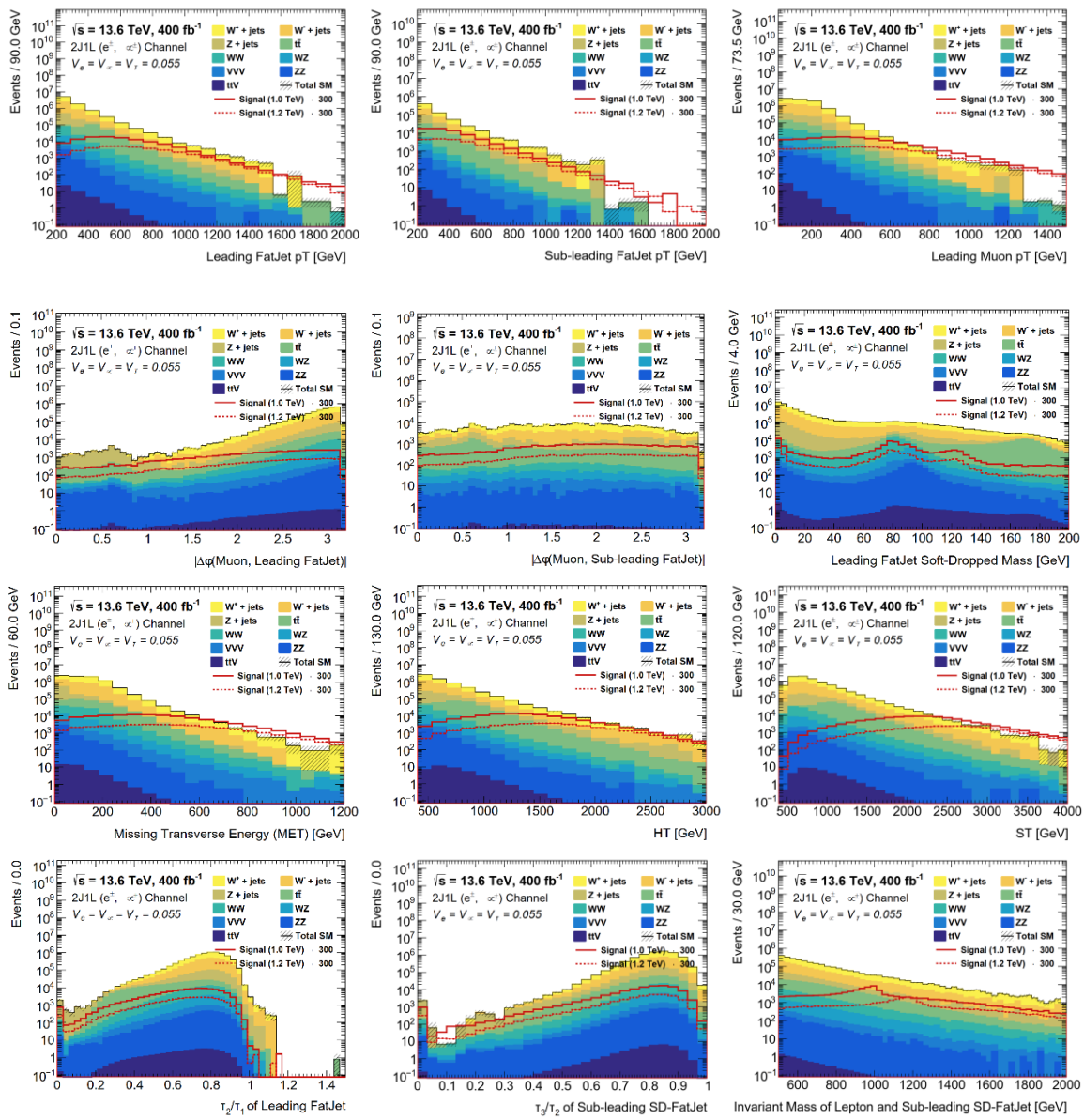
III. Multivariate Analysis

For event-level signal discrimination, we employ the **BDT** method from the **TMVA** package within the ROOT analysis framework. All event-level BDTs are initially configured with 400 trees, a maximum depth of 10, a minimum node size of 1.5% of the training sample, and 10 steps during node cut optimization. The AdaBoost algorithm with a learning parameter $\beta = 1.0$ is used for boosting. These hyperparameters will be further tuned to maximize background rejection for a given signal efficiency in the training samples. Additionally, the number of mass regions is included as a parameter and is expected to enhance the discrimination power of the BDTs. For each mass hypothesis, the output scores from all trained BDTs will be evaluated, and the BDT yielding the best expected exclusion limit will be selected for that mass point.

Analysis variables currently being used:

Object Kinematics (9):
$p_T(\ell_1),\ p_T(J_1),\ p_T(J_2),\ \tau_{21}(J_1),\ \tau_{32}(J_1),\ \tau_{21}(J_2),\ \tau_{32}(J_2), m_{SD}(J_1), m_{SD}(J_2)$
Angular kinematics (10):
$\Delta R(\ell_1,J_1)$, $\Delta R(\ell_1,J_2)$, $\Delta\phi(\ell_1,J_1)$, $\Delta\phi(\ell_1,J_2)$, $\Delta\phi(\ell_1,\vec{p}_T^{miss})$, $\Delta\phi(J_1,\vec{p}_T^{miss})$, $\Delta\phi(J_2,\vec{p}_T^{miss})$ $\lvert\Delta\eta(\ell_1,J_1)\rvert$, $\lvert\Delta\eta(\ell_1,J_2)\rvert$, $\Delta R(J_1,J_2)$
Event Kinematics (7):
E_T^{miss}, H_T, S_T, $M_{inv}\left(\ell_1,J_1^{(SD)}\right)$, $M_{inv}\left(\ell_1,J_2^{(SD)}\right)$, $M_{inv}\left(J_1^{(SD)},J_2^{(SD)}\right)$, $M_{inv}(system)$

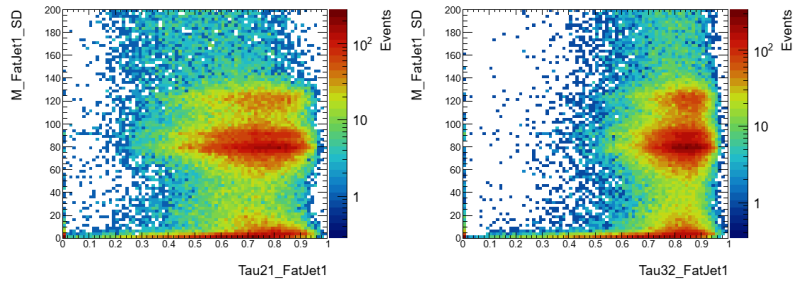
Each input variable will be studied to assess its contribution to the classifier's performance. This is essential for establishing a systematic understanding of the discriminating variables being used. Below are distributions of some analysis variables:



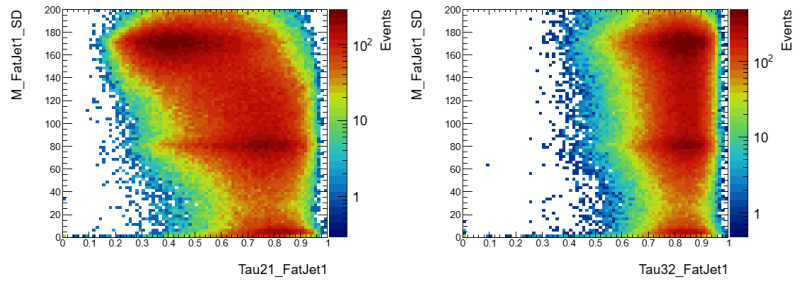
IV. Tagging Hadronically Decaying boosted EW and Higgs boson using Machine Learning Methods

This work is essential for identifying (tagging) final-state EW and Higgs bosons fatjets. A key challenge is distinguishing the signal fatjets from those originating in background processes, such as hadronic decays from top quark pairs ($t\bar{t}$) and diboson production. Applying effective tagging methods will be crucial for the event-level sensitivity. These methods are particularly important for enhancing the sensitivity of final states containing exactly two fatjets and zero leptons (2J0L Channel). This signature arises primarily from the channel channel $N^+N^- \rightarrow W^+\nu W^-\nu$, where the $W^\pm$ bosons decay hadronically. Due to its larger cross-section, this channel holds greater statistical significance than the trilepton (3L) and four-lepton (4L) channels. For this analysis, we employ machine learning-based tagging techniques. A classifier is trained using features derived from **N-subjettiness**, **energy**

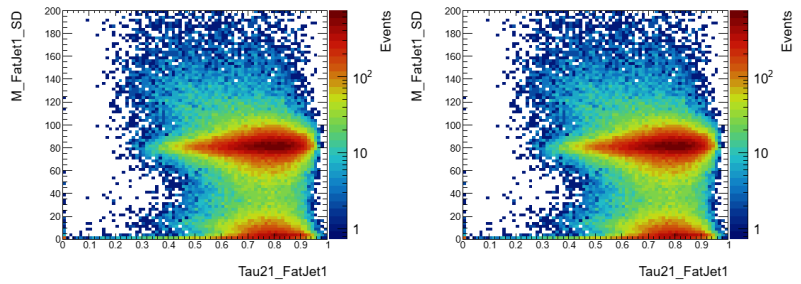
correlation functions, and the **kinematics of the fatjet constituents**. Below are some unweighted scattering plots of the 2 and 3-prongrness against the fatjet soft dropped mass:



Signal Events ($m_N = 1.0$ TeV). The Higgs fatjets are very obvious.



$t\bar{t}$ background events.



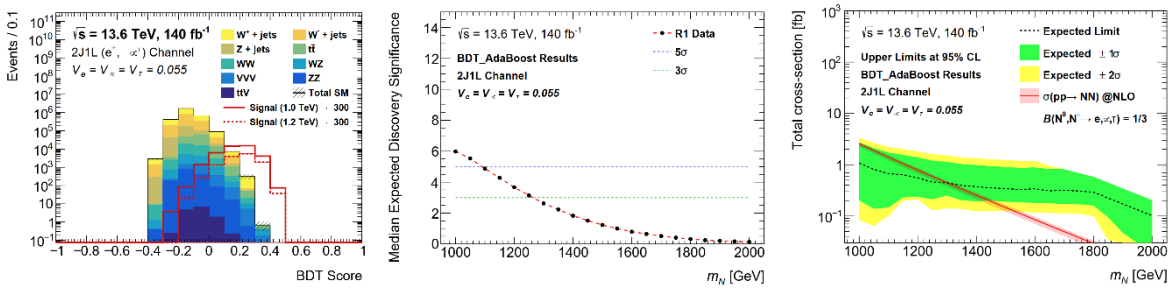
$W^{\pm}W^{\mp}$ background events

V. Statistical Analysis

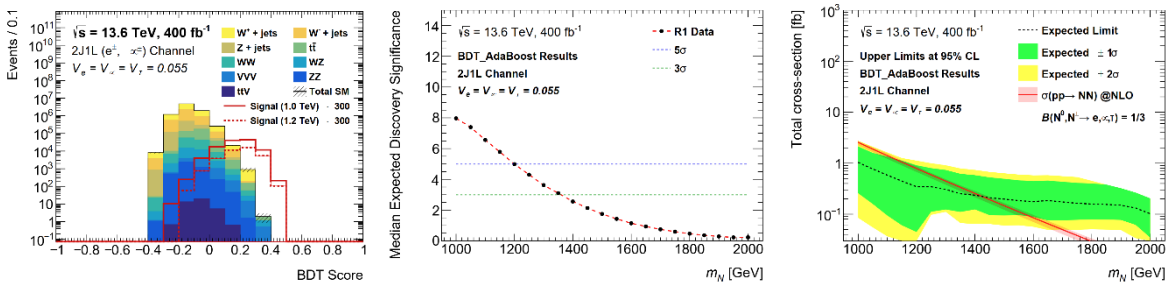
We use the **pyhf** python package for constructing the binned-likelihood model. Upper limits on μ are obtained using the CL_s method, for $CL_s = 0.05$ (95% CL). The discovery sensitivity is evaluated using the one-sided test statistic q_0.

五、预期成果

We use AdaBoosted BDTs distributions directly as the main discriminant for statistical analysis. Below we show the BDT Score distributions, derived significance and exclusion limits using 140 fb^{-1} of data:



As shown above, we project mass exclusion limits at 1.30 TeV, with integrated luminosities of 140 fb^{-1} with 95% confidence level. Below are the results projected at integrated luminosities of 400 fb^{-1}:



Here we project mass exclusions limits at 1.42 TeV with 95% confidence level. Additionally, according to our results, leptonic triplets with mass up to 1.2 (1.35) TeV could be discovered with 5σ (3σ) significance. Comparing our results to [7]:

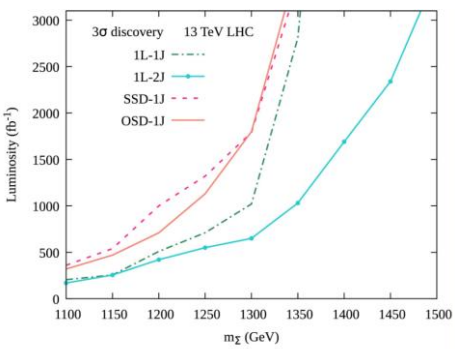


Fig. 5. Required Luminosity for 3σ discovery of the triplet fermions in different analysis channels.

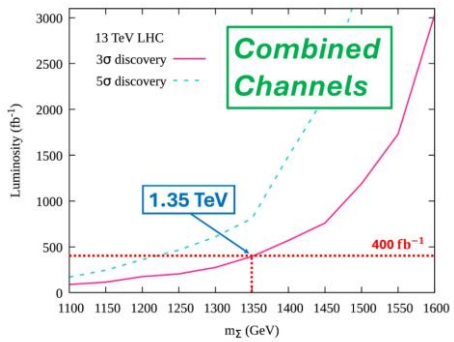


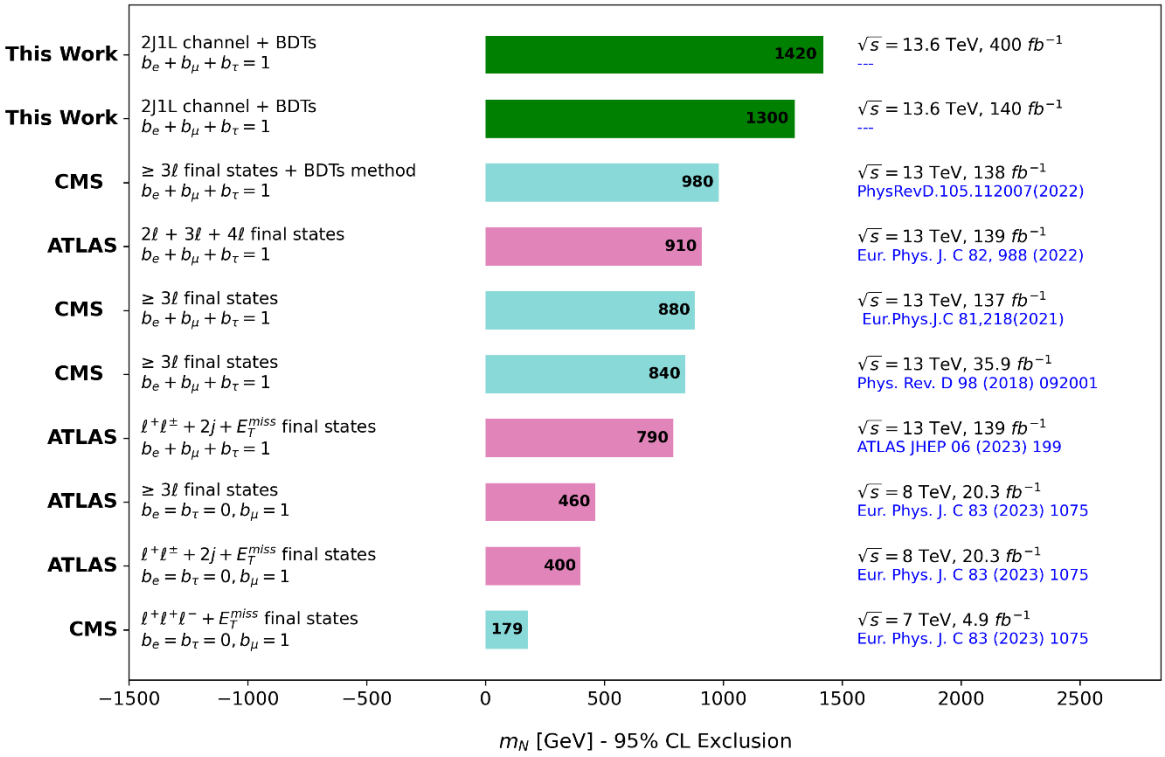
Fig. 6. Required Luminosity for 3σ and 5σ discovery of the triplet fermions in the combined channels.

triplets with mass up to 1265 (1380) and 1480 (1600) GeV could be discovered with 5σ (3σ) significance at 500 and 3000 fb^{-1} luminosities, respectively. Translating to 400 fb^{-1}, is 1.35 TeV with 3σ. Which is similar to our results **using the 2J1L channel alone**. Hence, we expect that with further application of the fatjet tagging ML results, and with including results of the remaining proposed channels, our mass exclusion limits can by increased by at least **2x higher**. With that being done, we also get:

1) A fully developed and optimized **multi-channel search strategy** based on final states containing multiple large-R jets and leptons, complete with object definitions, selection criteria,

and background estimation prescriptions.

2) A set of trained **Boosted Decision Tree (BDT) multivariate discriminants** for different signal mass hypotheses and decay scenarios, providing a performant and reusable analysis tool.
3) A complete **statistical analysis framework** using a profile likelihood ratio and the CLs method, incorporating main systematic uncertainties, ready to derive the final limits.
4) A comprehensive **journal article** detailing the analysis and results.
5) **Internal ATLAS notes** and presentations proposing this optimized search strategy to the relevant physics group, providing a direct blueprint for future data analysis.



六、已取得的与论文研究内容相关的成果

No articles are published yet.

七、研究计划与进度安排

Phase 1: Finalize Core Analysis & Produce Results (Next 2 Months)

- **Key Tasks:**
  1. **Complete Remaining Simulations:** Finalize any pending background Monte Carlo samples to ensure robust background modelling. Possibly variable weighting of the background samples will be applied to get a smoothly falling background distribution.
  2. **Lock Analysis Cuts & BDT Training:** Finalize the event selection criteria and complete the training/optimization of the Boosted Decision Trees for all defined channels.
  3. **Generate Final Distributions:** Produce the definitive signal and background yield tables, as well as all key kinematic plots/

Phase 2: Statistical Analysis & Uncertainty Finalization (Month 3)

- **Key Tasks:**
  1. **Incorporate Systematics:** Finalize the evaluation and implementation of major systematic uncertainties (e.g., jet energy scale, luminosity, theory cross-sections).
  2. **Extract Limits & Significance:** Calculate the final **95% CL exclusion limits** and **discovery significances** (3σ, 5σ) for the combined channels.

Phase 3: Thesis Writing & Integration (Months 4)

- **Key Tasks:**
  1. **Draft Core Chapters:** Concentrate on writing the results and discussion chapters, directly integrating the finalized plots and numbers.
  2. **Update Methodology:** Refine the methodology chapter based on the completed analysis work.
  3. **Final Compilation & Editing:** Assemble the full draft, perform thorough proofreading, and format according to submission guidelines.

研究生本人签名：_____

指导教师（主导师）签名：_____

2025 年 12 月 22 日

中国科学技术大学学术学位研究生学位论文开题报告评审表

姓　　名	AHMED MEDHAT MOSTAFA ELSAYE	学　　号	SL23004001	性　　别	男
所在院系	近代物理系	学科、专业	粒子物理与原子核物理		
研究方向	实验粒子物理	指导教师	吴雨生		
学生类别	☑ 学术学位硕士　　☐ 学术学位博士				
拟撰写的学位论文题目	Search for type III seesaw heavy leptons in pp collisions at √s = 13.6 TeV with the ATLAS detector				
支持论文研究的科研项目					
学位论文是否保密	1．不保密（　）			导师签字	
	2．保密　（　）　密级：绝密（　）、机密（　）、秘密（　）				

开题报告评审组成员名单

姓名	职称	工作单位	签名

指导教师意见：

指导教师（主导师）签字：

年　　月　　日

评审小组意见：（是否通过开题论证，是否需要修改等）

☐ 通过			☐ 不通过
☐ 优秀	☐ 良好	☐ 一般	

评审小组组长签字：

年　　月　　日